

Finite-Dimensional Targets for Banach-Valued Sobolev Extension Sets

FA/Banach run fa_banach_001

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Status

This packet records a `partial_result_likely_valid` for Question 1.1 of Garcia-Bravo-Ikonen-Zhu [1]. The result is narrow: it treats only finite-dimensional Banach targets. It does not solve the infinite-dimensional part of the question.

Source Question

The source question appears as Question 1.1 on PDF page 3 of [1].

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Question 1.1. *Let $1 < p < \infty$ and Z be a p -PI space. Are $W^{1,p}$ -extension sets $\Omega \subset Z$ also $\mathbb{V}$ -valued $W^{1,p}$ -extension sets for all Banach spaces $\mathbb{V}$?*

More generally, does there exist a Banach space $\mathbb{V}_0$ such that every $\mathbb{V}_0$ -valued $W^{1,p}$ -extension set is a $\mathbb{V}$ -valued $W^{1,p}$ -extension set for all Banach spaces $\mathbb{V}$?

Many of the known real-valued extension results generalize to the Banach-valued setting with relative ease when the real-valued assumption is replaced by the corresponding Banach-valued one. The main difficulty in Question 1.1 arises in connecting the extensions properties for different Banach spaces: For example, if Ω is a real-valued $W^{1,p}$ -extension set, an obvious approach to proving the existence of a, e.g. ℓ^∞ -valued $W^{1,p}$ -extension operator is to extend componentwise. However, without knowing that the extension operator commutes with projections, it is not obvious why the extension obtained in this way is even an $L^p(Z; \ell^\infty)$ -function. Even if $L^p(Z; \ell^\infty)$ -boundedness of the extension could be deduced (as one can do in many important cases, cf. Theorem 1.3 below), the Sobolev regularity of the extension is more difficult to deduce. We were able to make this idea work only in some special cases, cf. Section 1.4 below.

In words, for $1 < p < \infty$ and a p -PI space Z , the question asks whether every real-valued $W^{1,p}$ -extension set $\Omega \subset Z$ is automatically a $\mathbb{V}$ -valued $W^{1,p}$ -extension set for all Banach spaces $\mathbb{V}$. The source paper explains that extending coordinates is not obviously valid for infinite-dimensional targets such as ℓ^∞ , because the resulting map need not be controlled in $L^p(Z; \ell^\infty)$ or in Sobolev regularity.

Idea of the Proof

The infinite-dimensional obstruction is absent in finite dimension. After identifying the target with ℓ_∞^n , a vector-valued function has only finitely many coordinates. We extend each coordinate by the

given real-valued extension operator. The L^p norm of the recombined vector is bounded by the finite sum of the coordinate L^p norms, and the maximum of finitely many coordinate upper gradients is an upper gradient for the recombined vector. An arbitrary finite-dimensional Banach target is linearly isomorphic to ℓ_∞^n , so the result transfers with a constant depending on that isomorphism.

Theorem

Theorem 1. *Let $1 \leq p < \infty$, let Z be a metric measure space, and let $\Omega \subset Z$. Suppose that Ω is a real-valued $W^{1,p}$ -extension set: there is an operator*

$$E_{\mathbb{R}} : W^{1,p}(\Omega; \mathbb{R}) \rightarrow W^{1,p}(Z; \mathbb{R})$$

such that $E_{\mathbb{R}}f|_{\Omega} = f$ and

$$\|E_{\mathbb{R}}f\|_{W^{1,p}(Z)} \leq C_E \|f\|_{W^{1,p}(\Omega)}$$

for all $f \in W^{1,p}(\Omega)$. Then, for every finite-dimensional Banach space $\mathbb{V}$, Ω is a $\mathbb{V}$ -valued $W^{1,p}$ -extension set. More precisely, if $\dim \mathbb{V} = n$ and $A : \mathbb{V} \rightarrow \ell_\infty^n$ is a linear isomorphism, then there is an extension operator

$$E_{\mathbb{V}} : W^{1,p}(\Omega; \mathbb{V}) \rightarrow W^{1,p}(Z; \mathbb{V})$$

whose norm is bounded by a constant depending only on p , n , C_E , and $\|A\| \|A^{-1}\|$.

Proof. First assume $\mathbb{V} = \ell_\infty^n$. Let $u = (u_1, \dots, u_n) \in W^{1,p}(\Omega; \ell_\infty^n)$ and define

$$H = (E_{\mathbb{R}}u_1, \dots, E_{\mathbb{R}}u_n).$$

Since the coordinate projections $\ell_\infty^n \rightarrow \mathbb{R}$ are 1-Lipschitz and linear, each u_i lies in $W^{1,p}(\Omega)$ and

$$\|u_i\|_{W^{1,p}(\Omega)} \leq \|u\|_{W^{1,p}(\Omega; \ell_\infty^n)}$$

up to the harmless convention-dependent choice of equivalent Sobolev norm. Thus each coordinate extension belongs to $W^{1,p}(Z)$ and restricts to u_i on Ω .

Choose p -weak upper gradients g_i for $E_{\mathbb{R}}u_i$. Then

$$g = \max_{1 \leq i \leq n} g_i$$

is a p -weak upper gradient for H : for p -almost every rectifiable curve γ and all curve parameters a, b ,

$$|H(\gamma(a)) - H(\gamma(b))|_{\ell_\infty^n} = \max_i |E_{\mathbb{R}}u_i(\gamma(a)) - E_{\mathbb{R}}u_i(\gamma(b))| \leq \int_{\gamma|_{[a,b]}} g \, ds.$$

Moreover,

$$\|H\|_{L^p(Z; \ell_\infty^n)} \leq \left(\sum_{i=1}^n \|E_{\mathbb{R}}u_i\|_{L^p(Z)}^p \right)^{1/p}$$

and similarly

$$\|g\|_{L^p(Z)} \leq \left(\sum_{i=1}^n \|g_i\|_{L^p(Z)}^p \right)^{1/p}.$$

Taking the g_i nearly minimal and using the bound for $E_{\mathbb{R}}$ gives

$$\|H\|_{W^{1,p}(Z; \ell_\infty^n)} \leq C(p, n, C_E) \|u\|_{W^{1,p}(\Omega; \ell_\infty^n)}.$$

Thus Ω is an ℓ_∞^n -valued extension set.

For a general finite-dimensional $\mathbb{V}$, fix a linear isomorphism $A : \mathbb{V} \rightarrow \ell_\infty^n$. Given $u \in W^{1,p}(\Omega; \mathbb{V})$, apply the ℓ_∞^n construction to Au and set

$$E_{\mathbb{V}}u = A^{-1}E_{\ell_\infty^n}(Au).$$

Postcomposition with bounded linear maps preserves Newton–Sobolev membership and multiplies upper gradients by at most the operator norm; this is the standard Lipschitz postcomposition property used in [1]. Therefore $E_{\mathbb{V}}u \in W^{1,p}(Z; \mathbb{V})$, it restricts to u on Ω , and its norm is bounded by a constant depending only on p, n, C_E and $\|A\| \|A^{-1}\|$. $\square$

Verification Notes

The proof uses only finite coordinate recombination and the standard postcomposition property for Newton–Sobolev maps. No computational check is needed. The statement is consistent with the source paper’s discussion: the componentwise obstruction there concerns infinite-dimensional targets, where coordinatewise extensions may fail to define an L^p -controlled sequence-valued map.

Limitations And Search Bounds

This result leaves the main infinite-dimensional form of Question 1.1 open. It does not treat c_0 , ℓ^∞ , arbitrary separable targets, or Banach spaces with approximation properties. It also does not characterize which sets are real-valued extension sets.

The cheap run indexes were searched for 2208.12594, `Banach-valued Sobolev, finite-dimensional target, componentwise extension`, and `real-valued extension set`. A web/arXiv search on June 15, 2026 for finite-dimensional Banach-valued Sobolev extension variants found the source paper itself but no separate later paper recording this exact finite-dimensional target subcase.

Human Review Recommendation

Review as a modest finite-dimensional subcase. The main point to check is that the chosen convention for the $W^{1,p}$ norm and p -weak upper gradients matches the coordinate estimates above; under the standard equivalent Newton–Sobolev norms used in [1], the constants are harmless.

References

- [1] M. Garcia-Bravo, T. Ikonen, and Z. Zhu, *Extensions and approximations of Banach-valued Sobolev functions*, arXiv:2208.12594.